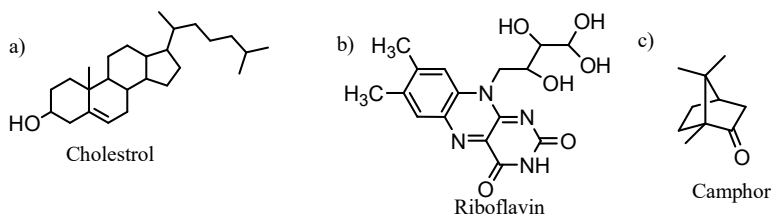


FOC work sheet -II

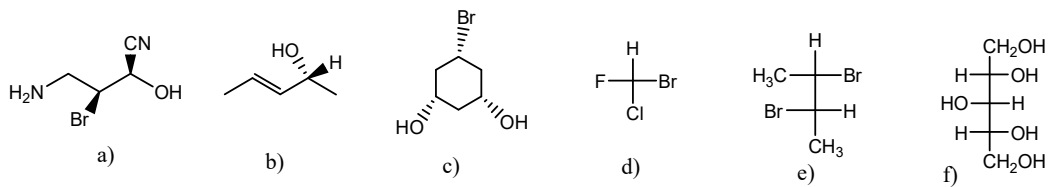
1. Draw a structural formula for an optically active compound with the molecular formula

- a. $C_4H_{10}O$ b. $C_5H_{11}Cl$ c. $C_4H_8(OH)_2$ d. C_6H_{12}

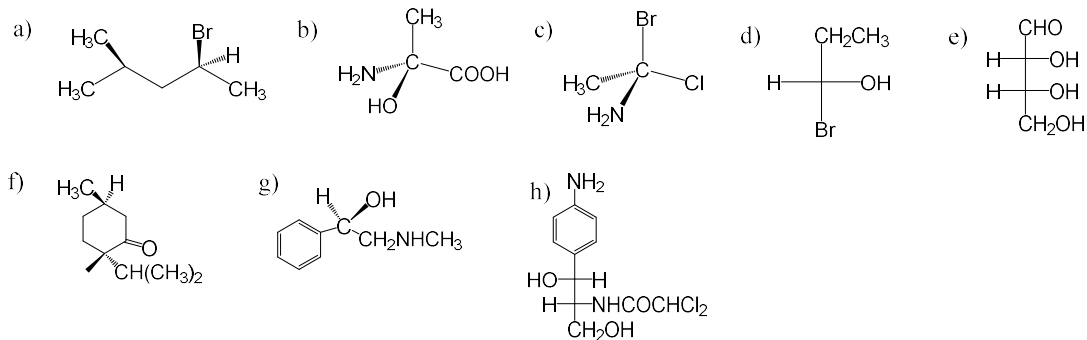
2. Which of the following molecules have chiral center? Place an asterisk to each chiral center.



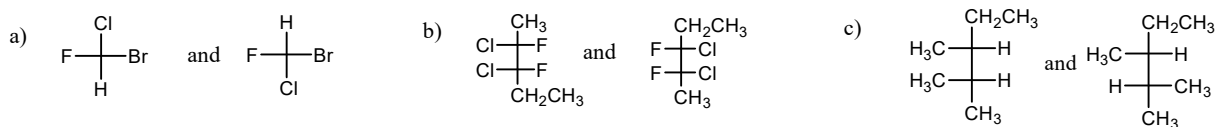
3. Label each molecule as chiral, achiral, or achiral /meso. Designate each chirality center as R or S



4. Assign R or S configuration for each chiral carbon.

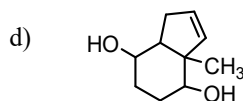
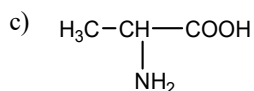


5. What is the relationship between each or the following pairs of compounds? (identical, enantiomers, diastereomers, none)



6. How many stereoisomers possible from each of the following molecules?

- a) 2,3,4- trihydroxypentanal b) 2-chlorocyclohexanol



7. How many stereoisomers are possible for each of the following structure? Draw them, and name each by R-S and E-Z conventions

a. 3-methyl-1,4-pentadiene

c. 2-bromo-5-chloro-3-hexene

b. 3-methyl-1,4-hexadiene

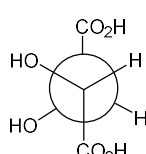
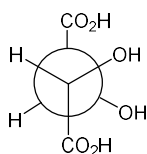
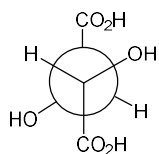
d. 2,5-dichloro-3-hexene

8. What is the difference between a) enantiomers and diastereomers?

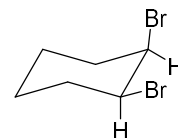
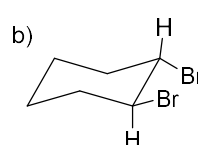
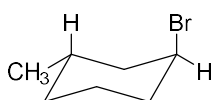
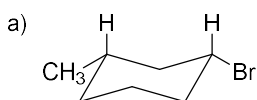
b) meso compounds and achiral compounds

9. Identify and draw the most stable conformation of a) ethane b) Butane c) Cyclohexane

10. Below are Newman projection for the three tartaric acids (R,R), (S,S), and *meso*. Which is which?



11. Name the following *cis-trans* pair.



12. Explain with the aid of conformational structure why *cis*-1,3-dimethylcyclohexane is more stable than *trans*-1,3-dimethylcyclohexane, whereas the reverse order of stability is observed for the 1,2 and 1,4 isomers.

13. Does ascorbic acid (vitamin C) has a D or L configuration?

